

PSA.

◦ TODAY'S LECTURE: HW 2, Q1-6

✓ Homework deadline HW2 → Tuesday 23:59

✓ MatLab assignments. → SUNDAY.

MATLAB $\begin{cases} 15pt \\ 20pt \end{cases}$

◦ Week 3.

HW - 30
30
10
15
15

Lecture 04:

1.9. Integrating Factors + Undetermined Coeffs.

3.6 Second order linear equations.

2.3. Harmonic oscillator

Ex. Suppose that a 400ml mug of coffee has 100ml of coffee and 4g of sugar.

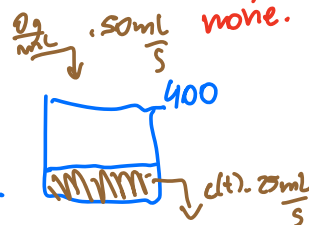
To fix that crime, non-contaminated coffee (0g/ml) is added at a rate of 50 ml per second.

Suppose that your worst enemy drinks 25 ml per second out of the well mixed mug. How many grams of sugar will be there by the time the mug is full? Hopefully none.

Use an integrating factor.

$$m' = \overset{\text{vin}}{0 \cdot 50} - \underset{\text{cin}}{c(t) \cdot 25} = -25 \cdot \frac{m}{100+25t} = \frac{-m}{4+t}$$

$$m(0) = 4$$



$$V(t) = 100 + 25t$$

$$m' + \frac{1}{4+t} m = 0$$

$$\mu(t) = e^{\int \frac{1}{4+t} dt} = e^{\ln|4+t|} = 4+t.$$

Multp. by $\mu(t)$:

$$(4+t)m' + 1 \cdot m = 0$$

prod. rule. $\Rightarrow ((4+t) \cdot m)' = 0 \Rightarrow (4+t) \cdot m = C$

integrate $\hookrightarrow m(t) = \frac{C}{4+t}$

$$\mu(t) (y' + p(t) \cdot y) = q(t) \mu(t)$$

$$\underbrace{\left(e^{\int p(t) dt} \cdot y \right)'} = q(t) \cdot \mu(t)$$

2nd order linear diff. equations. with constant coefficients.

equation: $ax'' + bx' + cx = 0$

$$y(t) = e^{\lambda t}, y'(t) = \lambda e^{\lambda t}, y''(t) = \lambda^2 e^{\lambda t}$$

$$bx' + cx = 0 \\ x' = -\frac{c}{b}x$$

Subst.

$$0 = a(\lambda^2 e^{\lambda t}) + b(\lambda e^{\lambda t}) + c(e^{\lambda t}) = \underbrace{e^{\lambda t}}_{\neq 0} [a\lambda^2 + b\lambda + c]$$

characteristic equation: $a\lambda^2 + b\lambda + c = 0$ λ_1, λ_2 roots.

general solution: $c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$

motivation: (linear independence)

$$\{v_1, v_2\} \quad v_1 \neq k \cdot v_2 \quad \lambda_1 \neq \lambda_2$$

$$e^{\lambda_1 t} \neq k \cdot e^{\lambda_2 t} \rightarrow \underbrace{e^{(\lambda_1 - \lambda_2)t}}_{e^{\lambda t}} \stackrel{?}{=} k$$

equality holds iff $\lambda_1 = \lambda_2$.

" If $\lambda_1 \neq \lambda_2$, then $y_1(t) = e^{\lambda_1 t}$, $y_2(t) = e^{\lambda_2 t}$ are l.i. solutions for your ODE and

$$y(t) = c_1 y_1 + c_2 y_2.$$

Example:

$$a) \quad y'' = -3y' + 4$$

$$y'' + 3y' - 4 = 0 \quad \rightarrow \quad \lambda^2 + 3\lambda - 4 = 0$$
$$\lambda_1 = -4 \quad \lambda_2 = 1.$$

$$y(t) = c_1 \cdot e^{-4t} + c_2 \cdot e^t.$$

$$b) \quad y'' - 5y' + 6 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0 \quad \rightarrow \quad \lambda_1 = 2$$
$$\lambda_2 = 3$$

$$y(t) = c_1 e^{2t} + c_2 e^{3t}$$

but ... what about complex roots? $\lambda = a \pm bi$

EF.

Euler's Formula.

$$e^{\lambda t} = e^{(a+bi)t} = e^{at} (\cos(bt) + i \sin(bt))$$

Ex:

$$e^{2-3i} = e^2 (\cos(-3) + i \sin(-3))$$

$$e^{5i} = \cos(5) + i \sin(5)$$

$$e^{\frac{\pi}{2}i} = \underbrace{\cos\left(\frac{\pi}{2}\right)}_0 + i \underbrace{\sin\left(\frac{\pi}{2}\right)}_1 = i$$

$$e^{\pi i} = \underbrace{\cos(\pi)}_{-1} + i \underbrace{\sin(\pi)}_0 = -1 \leadsto e^{\pi i} + 1 = 0$$

$$e^z = e^z.$$

Let $\lambda = a \pm bi$ be complex roots.

ex. $y'' - 4y' + 5y = 0$

ch. eq. $\lambda^2 - 4\lambda + 5 = 0$

$$\lambda = \frac{4 \pm \sqrt{16 - 20}}{2} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm 2\sqrt{-1}}{2} = 2 \pm i$$

complex-valued solutions: $e^{(2+i)t}$, $e^{(2-i)t}$

Fact: $e^{(2+i)t} = e^{2t} (\cos(t) + i \sin(t))$
 $+ e^{(2-i)t} = e^{2t} (\cos(-t) + i \sin(-t))$

$\cos(t) - i \sin(t)$
 $2e^{2t} \cos(t) \sim e^{2t} \cos(t)$

$\frac{e^{(2+i)t} - e^{(2-i)t}}{2i} = \cancel{e^{2t}} \sin(t)$

$\frac{e^{2t} \cos(t)}{e^{2t} \sin(t)} = \frac{\cos(t)}{\sin(t)} \neq k.$

Ex: $y'' - 4y' + 5y = 0$, $\lambda = 2 \pm i$

$y(t) = c_1 \cdot \operatorname{Re}(e^{\lambda t}) + c_2 \cdot \operatorname{Im}(e^{\lambda t})$

$y(t) = c_1 \cdot \underbrace{e^{2t} \cos(t)}_{\operatorname{Re}(e^{(2+i)t})} + c_2 \cdot \underbrace{e^{2t} \sin(t)}_{\operatorname{Im}(e^{(2+i)t})}$

$z = a + bi$

$\operatorname{Re}(z) = a$

$\operatorname{Im}(z) = b$

$z = \operatorname{Re}(z) + i \operatorname{Im}(z)$

$e^{at} \cos(bt) + i e^{at} \sin(bt)$

Ex: $y'' = -6y \leadsto \lambda^2 = -6 \Rightarrow \lambda = \pm \sqrt{6}i$

$$e^{\sqrt{6}it} = \cos(\sqrt{6}t) + i \sin(\sqrt{6}t)$$

$$y(t) = c_1 \cdot \cos(\sqrt{6}t) + c_2 \cdot \sin(\sqrt{6}t),$$

$-\sqrt{6}i$

$$y(t) = c_1 \cdot \cos(-\sqrt{6}t) + c_2 \cdot \sin(-\sqrt{6}t)$$

$$= c_1 \cdot \cos(\sqrt{6}t) - c_2 \sin(\sqrt{6}t)$$

Ex: $y'' - 6y' + 34 = 0$

$$\lambda^2 - 6\lambda + 34 = 0 \Rightarrow \lambda = \frac{6 \pm \sqrt{36 - 4 \cdot 34}}{2} = \frac{6 \pm \sqrt{36 - 136}}{2}$$

$$\lambda = \frac{6 \pm \sqrt{100}}{2} = \frac{6 \pm 10i}{2} = 3 \pm 5i$$

$$e^{(3+5i)t} = e^{3t} \cos(5t) + i e^{3t} \sin(5t)$$

$$y(t) = c_1 \cdot e^{3t} \cos(5t) + c_2 e^{3t} \sin(5t),$$

Repeated roots: a special case.

- Verify that $y_1(t) = Ce^{\lambda t}$ is a solution of

$$ay'' + by' + cy = 0$$

if λ is a double root.

$$y_1' = C\lambda e^{\lambda t}, y_1'' = C\lambda^2 e^{\lambda t}$$

$$a(C\lambda^2 e^{\lambda t}) + b(C\lambda e^{\lambda t}) + c(Ce^{\lambda t}) = Ce^{\lambda t}(\underbrace{a\lambda^2 + b\lambda + c}_{=0}) = 0.$$

- Motivation: $y_2(t) = ?$

$$\frac{y_2(t)}{y_1(t)} = \text{NOT A CONSTANT} = t \Rightarrow y_2(t) = te^{\lambda t}.$$

- HW 2, Q3: Verify that $y_2(t) = te^{\lambda t}$ is a solution too.

- Ex: $4y'' + 20y' + 25y = 0$

$$4\lambda^2 + 20\lambda + 25 = 0$$

$$(2\lambda + 5)^2 = 0 \Rightarrow \lambda = -\frac{5}{2}$$

$$y(t) = c_1 e^{-\frac{5}{2}t} + c_2 \cdot t e^{-\frac{5}{2}t}.$$

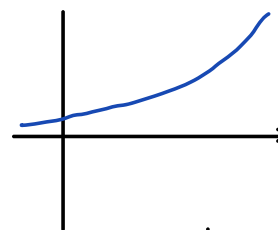
to do: * compare the solutions.

connect the corresponding cases:

pol 1. $\lambda^2 - 2\lambda + 2 = 0$
 $\Delta = 4 - 4 \cdot 2 = -4$

1. $y'' = 2y' - 2y$

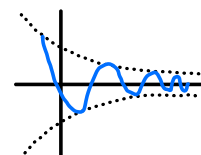
a) $y(t) = c_1 e^{1+it} + c_2 e^{1-it}$



pol 2. $\lambda^2 + 12\lambda + 36 = 0$

2. $y'' = -12y' - 36y$

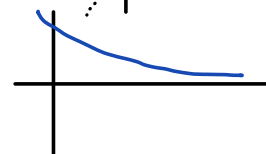
b) $y(t) = c_1 e^{-6t} \cos t + c_2 e^{-6t} \sin t$



3. $y'' = 3y'$

$\lambda^2 - 3\lambda = 0$
 $\lambda(\lambda - 3) = 0$

c) $y(t) = c_1 e^{0t} + c_2 t e^{0t}$



4. Why did the chicken cross the road? → d) Intermediate Value theorem

1. $y'' - 2y' + 2y = 0 \Rightarrow \lambda^2 - 2\lambda + 2 = 0$
 $\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 2}}{2}$
 $= \frac{2 \pm \sqrt{-4}}{2} = \frac{2 \pm 2\sqrt{-1}}{2} = 1 \pm i.$

2. $y'' + 12y' + 36y = 0 \Rightarrow \lambda^2 + 12\lambda + 36 = 0$
 $(\lambda + 6)^2 = 0$
 so $\lambda = -6.$

3. $y'' - 3y' = 0 \Rightarrow \lambda^2 - 3\lambda = 0 \Rightarrow \lambda(\lambda - 3) = 0$
 $\lambda = 0$ or $\lambda = 3.$

4. $y(1)$ 

 • $y(0)$

IVT: y cont. in $[0,1]$, $y(0) < \text{road} < y(1)$
 so there is $t_0 \in (0,1)$ s.t.
 $y(t_0) = \text{road}.$

Initial Value Problem

Two constants? Two conditions!

Ex. $y'' = 4y' + 5y$, $y(0) = 5$, $y'(0) = 1$.

$$\lambda^2 - 4\lambda - 5 = 0 \Rightarrow \lambda = 5 \text{ or } \lambda = -1$$

$$y(t) = c_1 e^{5t} + c_2 e^{-t}$$

$$y'(t) = 5c_1 e^{5t} - c_2 e^{-t}$$

$$y(0) = 5 = c_1 + c_2$$

$$y'(0) = 5c_1 - c_2 = 1$$

$$y(t) = e^{5t} + 4e^{-t}$$
$$\begin{cases} c_1 + c_2 = 5 \\ 5c_1 - c_2 = 1 \end{cases} \Bigg/ \begin{matrix} 6c_1 = 6 \\ \boxed{c_1 = 1} \end{matrix} \quad \begin{matrix} c_2 = 5 - 1 \\ \boxed{c_2 = 4} \end{matrix}$$

Ex. $y'' - 8y' + 20y = 0$, $y(0) = 0$, $y'(0) = 2$.

$$\lambda = 4 \pm 2i \Rightarrow y(t) = c_1 e^{4t} \cos(2t) + c_2 e^{4t} \sin(2t)$$

$$y'(t) = 4c_1 e^{4t} \cos(2t) + c_1 e^{4t} 2(-\sin(2t)) + 4c_2 e^{4t} \sin(2t) + c_2 e^{4t} 2 \cos(2t)$$

$$y'(0) = 4c_1 \underbrace{e^0 \cos(0)}_1 - 2c_1 \underbrace{e^0 \sin(0)}_0 + 4c_2 \underbrace{e^0 \sin(0)}_0 + 2c_2 \underbrace{e^0 \cos(0)}_1$$

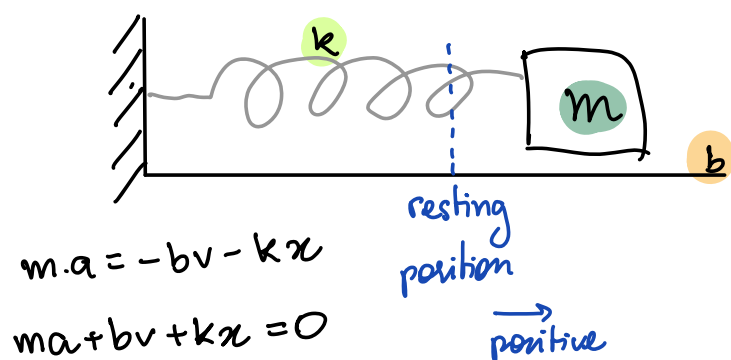
$$= 4c_1 + 2c_2 = 2.$$

$$y(0) = c_1 e^0 \cos(0) + c_2 e^0 \sin(0) = c_1 = 0. \quad \nearrow c_2 = 1$$

$$y(t) = e^{4t} \sin(2t).$$

Harmonic Oscillator

An object of mass m attached to a spring with spring constant k on a surface with friction b .



Let $x(t)$ be the distance from the resting position.

$$mx'' + bx' + kx = 0$$

$$m, b, k > 0$$

⊕ friction or damping coefficient.

◦ Undamped oscillator ($b=0$)

$$x'' + \frac{k}{m}x = 0 \quad \Rightarrow \quad \lambda^2 + \frac{k}{m} = 0$$

$$\lambda = \pm \sqrt{\frac{k}{m}} i$$

$$x(t) = c_1 \cos\left(\sqrt{\frac{k}{m}} t\right) + c_2 \sin\left(\sqrt{\frac{k}{m}} t\right)$$

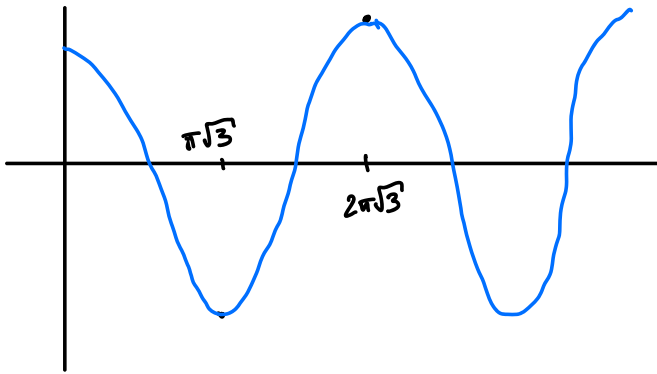
◦ Natural Frequency: $\omega = \sqrt{\frac{k}{m}}$

◦ Period of solution: $T = \frac{2\pi}{\omega}$

Ex. Object released at rest 3m from resting position,

mass 6kg, spring constant 2kg/s^2 .

hw: check $b=0$, $\lambda \in \mathbb{C}$, $\text{Re}(\lambda)=0$.



Damped oscillator ($b \neq 0$)

$$\lambda^2 + \frac{b}{m}\lambda + \frac{k}{m} = 0 \Rightarrow \lambda = \frac{-\frac{b}{m} \pm \sqrt{\frac{b^2}{m^2} - \frac{4k}{m}}}{2}$$
$$= \frac{-b}{2m} \pm \frac{1}{2m} \sqrt{b^2 - 4km}$$

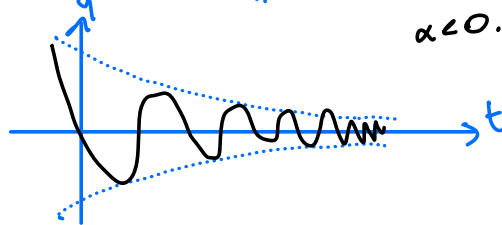
$\left\{ \begin{array}{l} \text{two distinct roots : } b^2 - 4km > 0 \\ \text{one double root : } b^2 - 4km = 0 \\ \text{two complex roots : } b^2 - 4km < 0 \end{array} \right.$

- Ⓘ $b^2 - 4mk < 0$: underdamped (complex)
- Ⓜ $b^2 - 4mk > 0$: overdamped. (real distinct)
- Ⓢ $b^2 - 4mk = 0$: critically damped (double root)

① $b^2 - 4mk < 0$: underdamped

$$\lambda = \frac{1}{2m}(-b \pm \sqrt{b^2 - 4mk}) = \alpha \pm \beta i$$

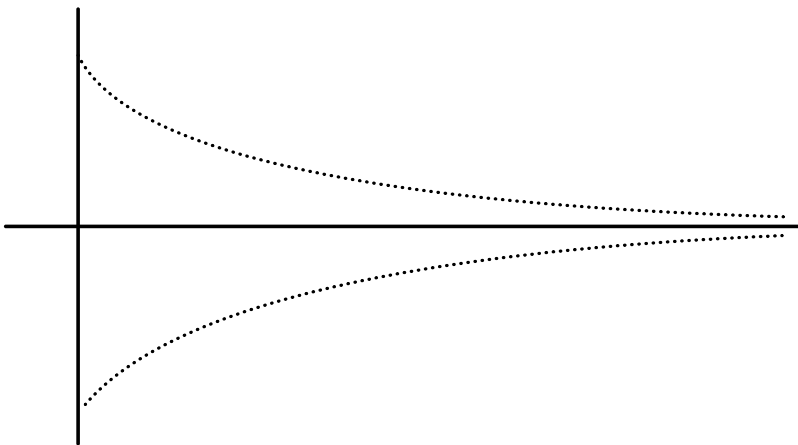
solution: $y(t) = c_1 e^{\alpha t} \cos(\beta t) + c_2 e^{\alpha t} \sin(\beta t)$
 $\alpha < 0.$



Ex: $m = 2 \text{ kg}$, $b = 8 \text{ kg/s}$, $k = 12.5 \text{ kg/s}^2$.

at rest, but 10m from starting position.

hw: check that it is underdamped.
plot a solution below.



① Overdamped: $b^2 - 4km > 0$.

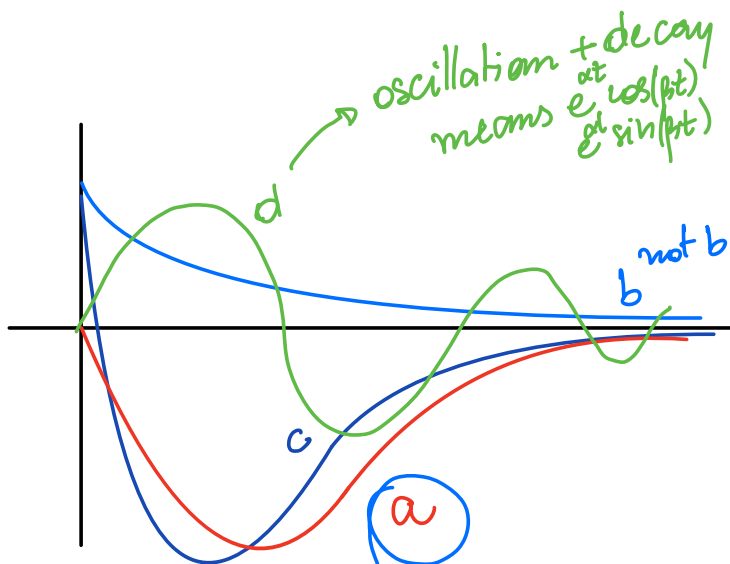
$$\lambda = \frac{-b \pm \sqrt{b^2 - 4km}}{2m}$$

$$\lambda_1 = \frac{-b}{2m} + \frac{\sqrt{b^2 - 4km}}{2m}$$

$$\lambda_2 = \frac{-b}{2m} - \frac{\sqrt{b^2 - 4km}}{2m}$$

• What happens with solutions?

$$y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$



Ex: $m = 1\text{kg}$, $b = 5\text{kg/s}$

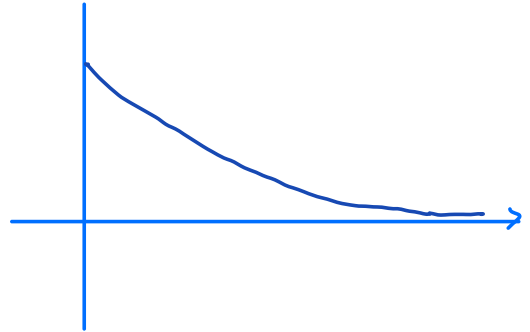
$k = 6\text{kg/s}^2$ initially at equilibrium but with velocity of 4m/s towards the spring.

but both a, c are overdamped profiles/solutions.

③ Critically damped: $b^2 - 4km = 0$.

root:
 $\lambda = \frac{-b}{2m}$

solution:
$$x(t) = c_1 e^{\frac{-b}{2m}t} + c_2 t e^{\frac{-b}{2m}t}$$
$$= e^{\frac{-b}{2m}t} (c_1 + c_2 t)$$



Question: why critically damped?

• does it change sign as t increases?

Ex: $m = 1$, $b = 5$, $k = \frac{25}{4}$, $x(0) = 1$, $x'(0) = -\frac{3}{2}$



Just check.

- HW 2: 1, 2, 3, 4, 5, 6, 15a (int. factor, und. coef.,
- Postclass 4 (shouldn't be too hard) 2nd order linear
+ harmonic oscillator)
- Preclass 5 (easy: intro to systems)

Wed, 6/15. Office Hours: 4:40pm - 5:40pm

Friday + Monday: HW2.

